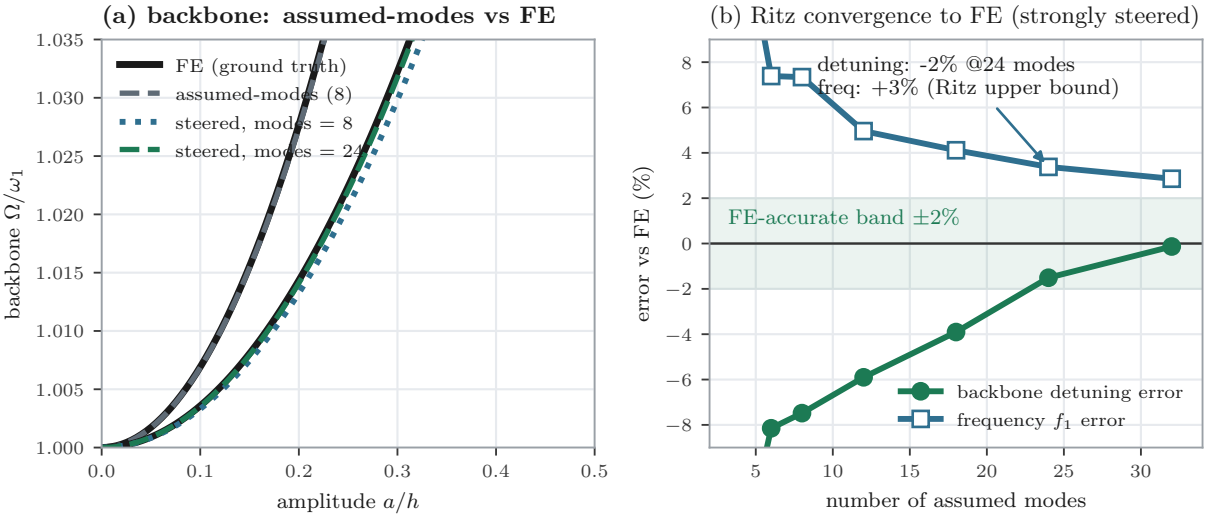

(a) backbone: assumed-modes vs FE
FE (ground truth)
assumed-modes (8)
steered, modes = 8
steered, modes = 24
backbone Ω/ω_1
amplitude a/h
1.035
1.030
1.025
1.020
1.015
1.010
1.005
1.000
0.0
0.1
0.2
0.3
0.4
0.5
(b) Ritz convergence to FE (strongly steered)
detuning: -2% @24 modes
freq: +3% (Ritz upper bound)
FE-accurate band ±2%
backbone detuning error
frequency f_1 error
error vs FE (%)
number of assumed modes
8
6
4
2
0
−2
−4
−6
−8
5
10
15
20
25
30